

When Does a Small Model Learn Portuguese?

Run ID: report_20260620T064400Z_final_training_20260620T053855Z_batch16

Status: empirical / methodological. The report does not make a formal SLT phase transition or causal claim, and it does not claim a changepoint in an SLT-derived local geometric estimate aligned with a behavioral transition. It reports behavior, LLC diagnostics, controls, and limitations from the approved real run directories only.

Headline: structured Portuguese behavior improved early in validation BPB, but the grammar probe is small and noisy; the primary LLC trajectory is non-monotone after excluding the token-0 diagnostic rejection. The evidence supports a reproducible diagnostic package, not a controlled aligned-changepoint result.

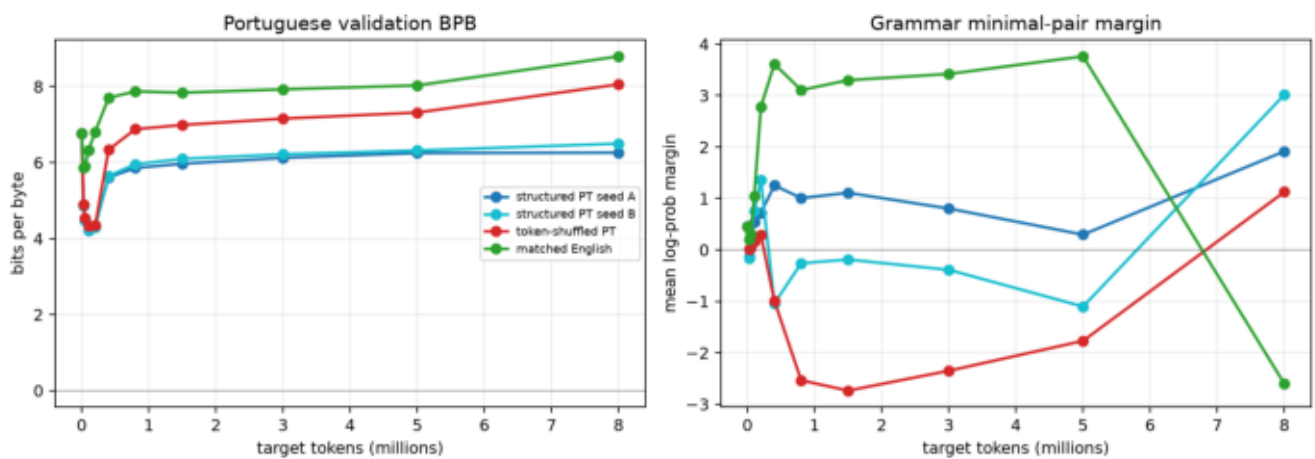
Design And Controls

Full-parameter FP32 next-token continued pretraining was run on TinyStories-8M. The real conditions are structured Portuguese seed A, structured Portuguese seed B, token-shuffled Portuguese, and matched English. Behavior was evaluated at every checkpoint. LLC was sampled only for structured Portuguese seed A at the frozen behavior-selected token subset.

The sampler used one global FP32 full-parameter SGLD configuration across selected checkpoints: 3 chains, 200 burn-in steps, 100 draws, 2 steps between draws, batch size 16, fixed reference set, and fixed sequence length 128.

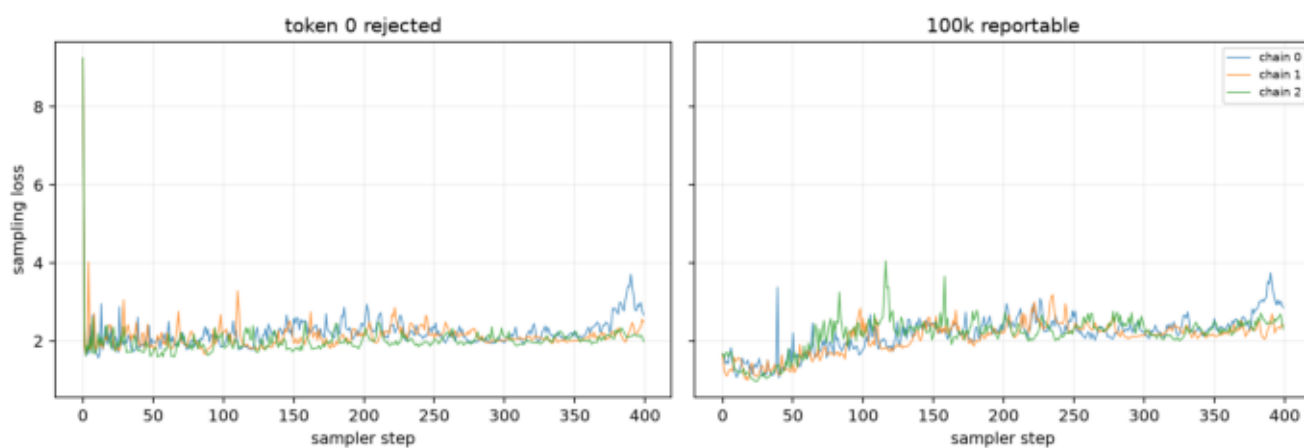
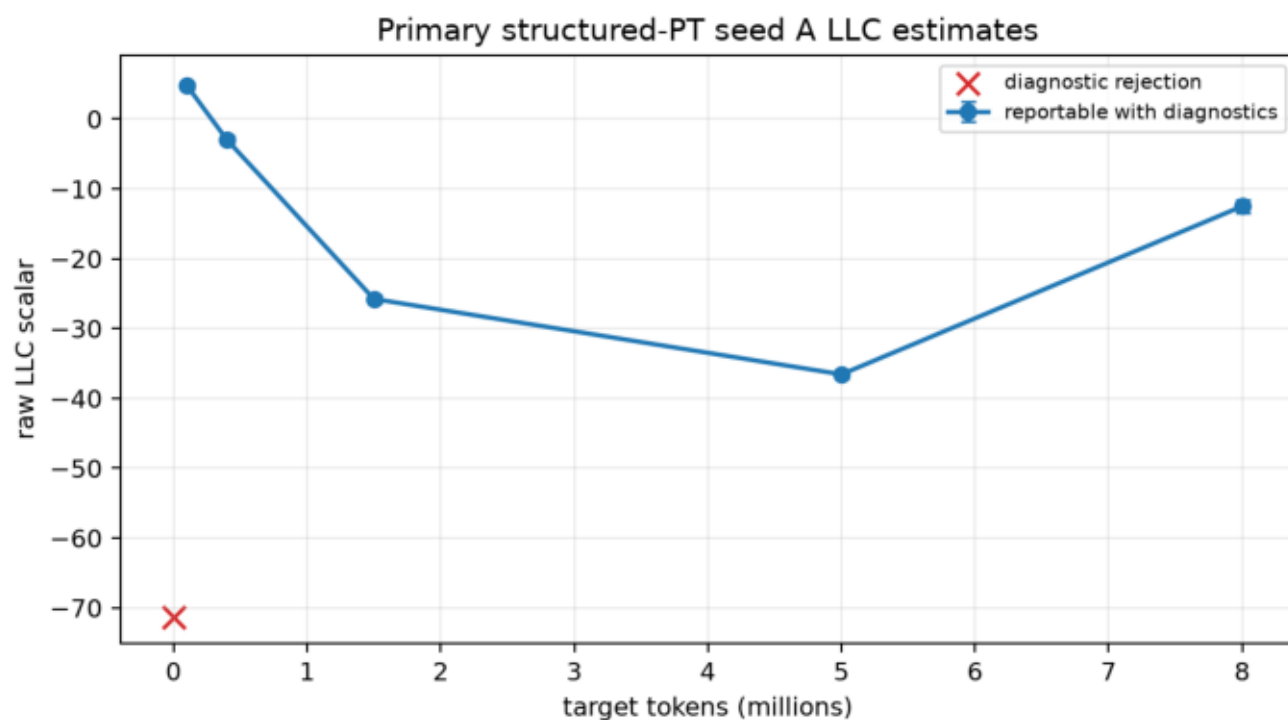
The grammar sanity check did not fully pass, and the report treats grammar margins as a small constructed diagnostic rather than a high-confidence capability measurement.

Behavior Trajectories



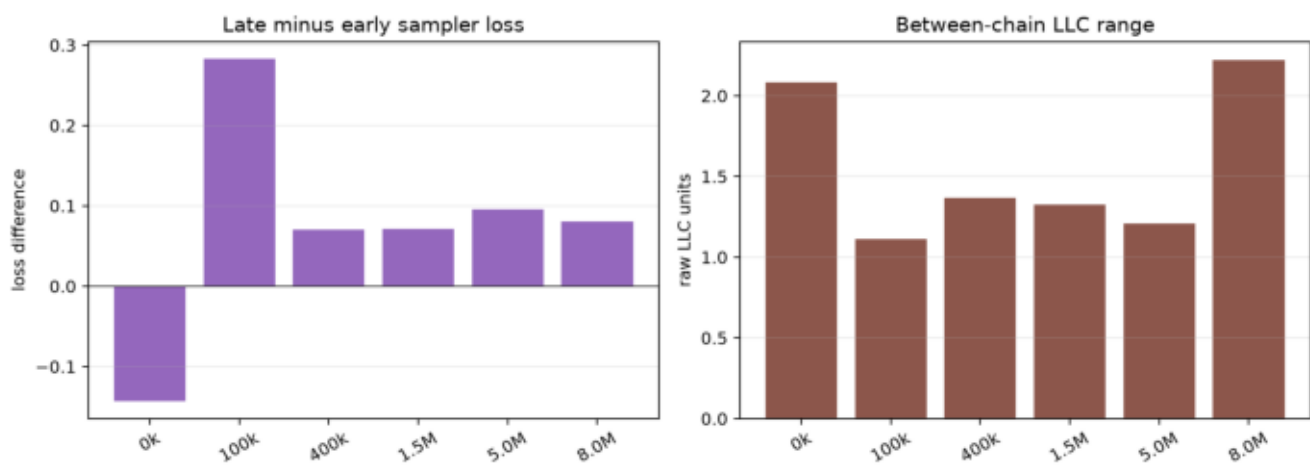
Controls are plotted in the same axes. Source data: [source_tables/behavior_trajectory_source.csv](#).

LLC Trajectory And Chain Diagnostics



Token 0 is shown as a rejected diagnostic point. The 100k trace is included as a representative reportable checkpoint. Full raw zarr traces, running estimates, and displacement traces are preserved in the LLC campaign run directory.

Sampler Summary



Limitations And Reproducibility

Main limitations: no LLC controls, no second-seed LLC, token-0 LLC rejection, tiny grammar probe, failed shared-prefix checks for two grammar items, repeated small split cycling, checkpoint-spacing uncertainty, and sampler/reference-set dependence.

Runtime and cost are recorded in `source_tables/runtime_cost_gates.csv`. Reproducibility metadata and hashes are recorded in `source_tables/reproducibility.csv` and `source_tables/selected_checkpoint_hashes.csv`. Every displayed table cell is covered by `validation/table_cell_verification.csv` and `validation/validation_summary.json`.